

Плест откометна правдоподобна

$$2 \cdot (\ln L_{UR} - \ln L_R) \xrightarrow{H_0} \chi^2_2$$

2 - число степеней свободы

$2 \cdot \ln \frac{L_{UR}}{L_R}$

Упражнение

$$X_1^T, \dots, X_{n_T}^T \sim \text{iid Poiss}(\lambda_T)$$

$$H_0: \lambda_T = \lambda_c$$

$$X_1^c, \dots, X_{n_c}^c \sim \text{iid Poiss}(\lambda_c)$$

$$H_A: \lambda_T > \lambda_c$$

$$L_{UR} = \mathbb{P}(X_1^T) \cdot \mathbb{P}(X_2^T) \cdot \dots \cdot \mathbb{P}(X_{n_T}^T) \cdot \mathbb{P}(X_1^c) \cdot \dots \cdot \mathbb{P}(X_{n_c}^c)$$
$$\frac{e^{-\lambda_T} \cdot \lambda_T^{x_i^T}}{x_i^T!} \cdot \dots \cdot \frac{e^{-\lambda_c} \cdot \lambda_c^{x_i^c}}{x_i^c!} \cdot \dots$$

$$L_{UR} \propto e^{-n_T \lambda_T} \cdot \lambda_T^{\sum x_i^T} \cdot e^{-n_c \lambda_c} \cdot \lambda_c^{\sum x_i^c}$$

$$\ln L_{UR} = -n_T \lambda_T + \sum x_i^T \cdot \ln \lambda_T - n_c \lambda_c + \sum x_i^c \cdot \ln \lambda_c \rightarrow \max_{\lambda_c, \lambda_T}$$

$$\hat{\lambda}_T = \bar{x}^T$$

$$\hat{\lambda}_c = \bar{x}^c$$

$$L_R \propto e^{-(n_T + n_c) \cdot \lambda} \cdot \lambda^{\sum x_i^T + \sum x_i^c}$$

$$\ln L_R = -(n_T + n_c) \cdot \lambda + \ln \lambda \cdot (\sum x_i^T + \sum x_i^c) \rightarrow \max_{\lambda}$$

$$\hat{\lambda} = \frac{1}{n_T + n_c} (\sum x_i^T + \sum x_i^c) = \bar{s}$$

$$LR = 2 \cdot \ln \frac{L_{UR}}{L_R} = 2 (\ln L_{UR} - \ln L_R) =$$

$$= 2 \cdot \left[(\sum x_i^T + \sum x_i^c) (1 - \ln \bar{s}) - \sum x_i^T (1 - \ln \bar{x}^T) - \sum x_i^c (1 - \ln \bar{x}^c) \right]$$
$$\sim \chi^2_1$$

- ① LR_{obs}
- ② $LR_{cr.}$
- ③ LR vs $LR_{cr.}$

Упражнения

	✓	✗
и		
н		

способ 1:

$$PF = \sum_{ij} \frac{(h_{ij} - \frac{n_{i.} \cdot n_{.j}}{n})^2}{h_{i.} \cdot h_{.j}}$$

$$\xrightarrow{d} \chi^2_{r \cdot c - 1}$$

Пирсон

1900 - 1915

$$\frac{h_{ij}}{n} \xrightarrow{p} p_{ij}$$

$$\xrightarrow{d} \chi^2_{(r-1)(c-1)} \text{ Фишер}$$

способ 2:

	p_{11}	p_{12}
	p_{21}	$1 - p_{11} - p_{12} - p_{21}$

$$n_{11} \quad n_{12}$$

$$n_{21} \quad n_{22}$$

$$n_{11} + n_{12} + n_{21} + n_{22} = n$$

$$\ln L_{ur} = \ln \left(\frac{n!}{n_{11}! n_{12}! n_{21}! n_{22}!} p_{11}^{n_{11}} p_{12}^{n_{12}} p_{21}^{n_{21}} (1 - p_{11} - p_{12} - p_{21})^{n_{22}} \right)$$

$$\max_{p_{11}, \dots} \ln L_{ur}$$

$$\hat{p}_{11} = \frac{n_{11}}{n}, \quad \hat{p}_{12} = \frac{n_{12}}{n}, \dots$$

$$\sum_{ij} \ln \left(\frac{h_{ij}}{n} \right)^{n_{ij}}$$

$$\ln L_R$$

$$P_{11} = P_{1\cdot} \cdot P_{\cdot 1}$$

$$\gamma \cdot w$$

non	x	m
	γ	$1-\gamma$

рас.	V	X
	w	$1-w$

Если 2 верты в независимости:

γw	$\gamma(1-w)$
$(1-\gamma)w$	$(1-\gamma)(1-w)$

$$\ln L_R = \ln \left(\frac{n!}{n_{11}! \dots n_{22}!} \right)$$

$$(\gamma w)^{n_{11}} \cdot (\gamma(1-w))^{n_{12}} \cdot ((1-\gamma)w)^{n_{21}} \cdot ((1-\gamma)(1-w))^{n_{22}}$$

Решаем где
кез. заданы
т.к. предполо-
жили кез-тб
выборки

$$\ln L_R \propto \ln \gamma^{(n_{11}+n_{12})} \cdot (1-\gamma)^{(n_{21}+n_{22})} \cdot w^{n_{11}+n_{21}} \cdot (1-w)^{n_{12}+n_{22}}$$

$$\hat{\gamma} = \frac{n_{11}+n_{12}}{n} \quad \hat{w} = \frac{n_{11}+n_{21}}{n}$$

$$LR = 2 \left(\ln L_{UR} - \ln L_R \right) \xrightarrow{H_0} \chi^2_q$$

$$2 \cdot \left(\sum_{ij} \ln \left(\frac{n_{ij}}{n} \right)^{n_{ij}} - \sum_i \ln \left(\frac{n_{i\cdot}}{n} \right)^{n_{i\cdot}} - \sum_j \ln \left(\frac{n_{\cdot j}}{n} \right)^{n_{\cdot j}} \right)$$

$$n_{i\cdot} = n_{i1} + n_{i2}$$

$$\xrightarrow{H_0} \chi^2_q$$

$$UR: r \cdot c - 1$$

$$R: (r-1)(c-1)$$

$$q = rc - 1 - (r-1) - (c-1) = rc - r - c + 1 = (r-1)(c-1)$$

$$\text{Plim}_{n \rightarrow \infty} \frac{LR}{PF} = \text{const} \quad \text{в предположении, что } H_0 \text{ верна}$$